



Make Your Own Nifty Night Lamp

T.K. Hareendran

This simple circuit can be used to light a small area at night. It is useful as a nifty bedroom night lamp, but it can also be used for other applications. The circuit is versatile and fully-automatic but draws very little electric current from the grid. Proposed enclosure of the project is shown in Fig. 1.

The circuit diagram of the nifty night lamp is shown in Fig. 2. It is built around two 47-ohm resistors (R1 and R2), rectifier diode 1N4007 (D1), 0.47 μ F, 400V capacitor (C2), transistor SS9014 (T1), LDR (LDR1), blue LED (LED1) and a few other components. CON1 is the input connector

for 230V AC mains supply.

This lamp is designed for 230V AC mains transformer-less power supply. The components used here would help maintain overall efficiency and safety. The capacitive transformer-less power supply is a low-cost alternative to a traditional power supply.

The constant-current configuration used here provides about 30mA at 5V DC. Capacitor C2 is the key component here, while resistors R1 and R2 limit inrush current. Values of R1 and R2 are chosen such that these do not dissipate too much power yet are large enough to limit inrush current.

Resistor R3 (470-kilo-ohm) in parallel with C2 creates a filter that will attenuate EMI travelling back to the grid. It also works as a bleeder resistor to discharge the capacitor when disconnected from AC mains (wall plug).

Light resistance of GL5528 LDR at 10 lux (at 25°C) is 8k Ω to 20k Ω , and it shoots up to one-mega-ohm at near 0 lux (see Fig. 3). So, when the night lamp is kept in the dark, T1 will be switched off and blue LED will glow. Note that the orthodox current-limiter resistor for LED is not used here.

As per calculations, power supply configuration will give a guaranteed output current close to 30mA at 5V, but exact output depends on tolerances of the components used.

Construction and testing

An actual-size PCB layout of the night lamp is shown in Fig. 4 and its components layout in Fig. 5. After assembling the circuit on the PCB,

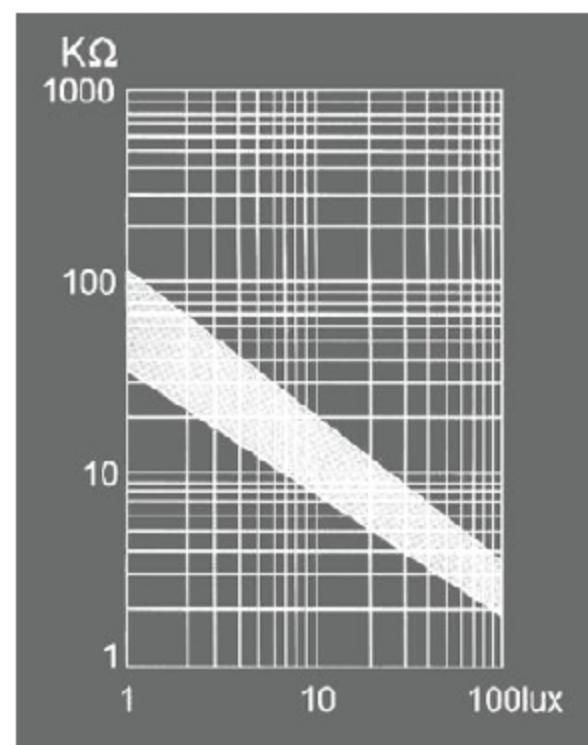


Fig. 3: Luminance versus photo resistance of the light sensor

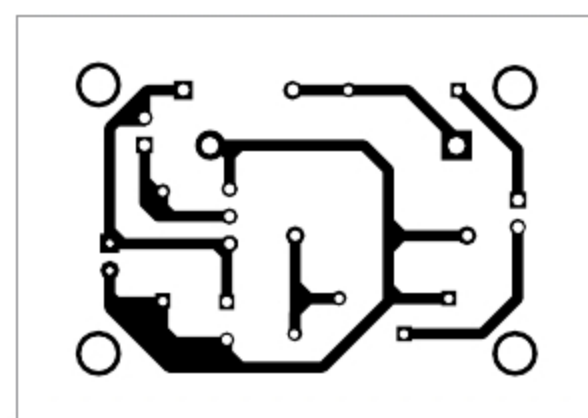


Fig. 4: Actual-size PCB layout of the nifty night lamp

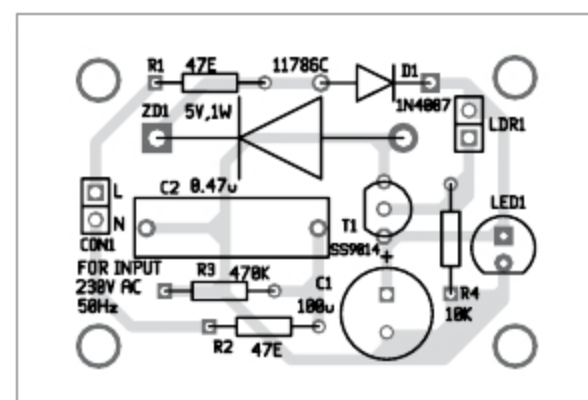


Fig. 5: Components layout for the PCB



Fig. 1: Proposed enclosure of the nifty night lamp

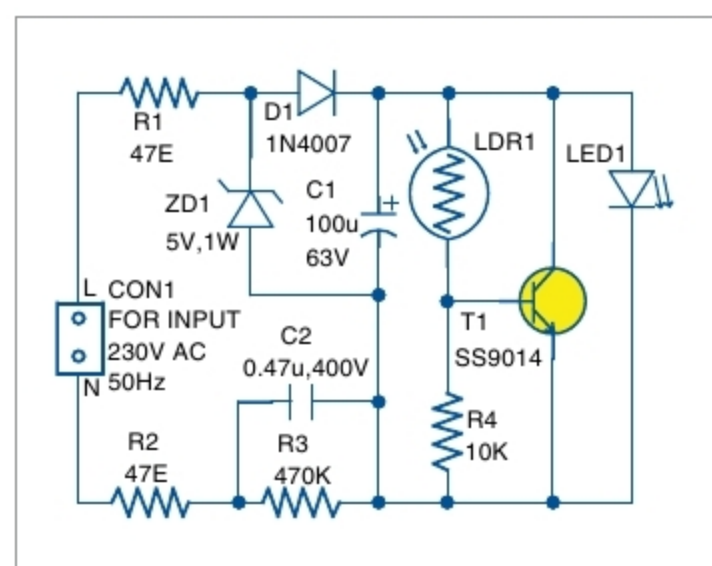


Fig. 2: Circuit diagram of the nifty night lamp